EXAM NUMBER: Personal FRED

KMTC/QP-08/TIS

Kenya Medical Training College

Department of Medical Laboratory Sciences

Final/Special/Supplementary Qualifying Examinations 2018

Diploma in Medical Laboratory Sciences (DMLS)

Paper: 05-A Immunology (100 marks)

Date:	Day:	Time:
14/02/2018	Wednesday	8.30 - 10.30 am

Instructions

1. Mobile phones are not allowed in the examination room
2. Write your exam number on the answer book used.
3. Do not write your name on the answer book.
4. Attempt both sections, I and II.

Section I comprises of M.C.Q.

5. Write MCQ answers in capitals on the first page of the answer book.
6. Be careful, as 1/2 mark will be deducted for every wrong response.

Section II comprises of Essay Questions.

7. Attempt all the questions. Make sure you understand the question before answering.

(3) Immunodeficiency (4) Hypersensitivity
(2) Immunological tolerance
(1) How cell mediated immunity help in immune response (20 marks)
MHC

INDEX NUMBER.....

SECTION I: (MCQ 40 MARKS)

1) The discovery of major histocompatibility was attributed to:-
   a) Edward Jenner
   b) Frank Burnet
   c) Elie Metchnikoff
   d) Jean Dausset
   e) Robert Koch

2) Lymphocytes are activated by antigen in the:-
   a) Blood stream
   b) Bone marrow
   c) Liver
   d) Lymph nodes
   e) Skin

3) The adaptive immune system has the following characteristics except:-
   a) Non specificity
   b) Self-tolerance
   c) B and T cell proliferate after activation
   d) Immunological memory
   e) Diversity of lymphocytes and receptors

4) T-Lymphocytes undergo differentiation in their subtypes in the:-
   a) Bone marrow
   b) Spleen
   c) Thymus
   d) Medulla
   e) Lymph nodes

5) The antibody isotype most commonly used for antibody replacement therapy is:-
   a) IgM
   b) IgD
   c) IgA
   d) IgE
   e) IgG

INDEX NUMBER.....

6) A synonym for an antigenic determinant is:-
   a) Immunogen
   b) Epitope
   c) Paratope
   d) Polysaccharide
   e) Hapten

7) Structure associated with specificity in acquired immunity include:-
   a) PAMP → Pathogen [illegible] molecular patterns
   b) BCR → B-cell Antigen receptor [illegible]
   c) TCR → T cell Antigen receptor
   d) TLR → Toll-like receptors
   e) PRR → Pattern recognition receptors

8) Cytotoxic T. cell recognizes antigen in association with:-
   a) Class II MHC determinant
   b) Class I MHC determinant
   c) Class III MHC determinant
   d) Class IV MHC determinant
   e) Class I and II MHC determinant

9) Which of the following statement(s) is/are true about T-cells;
   a) Are responsible for antibody mediated immunity
   b) Originate in the spleen
   c) Produce antibodies when activated
   d) Causes cell lysis
   e) Mature in the thymus

10) B-cells are characterized by:-
   a) their production in the thymus
   b) Release of cytotoxins
   c) Cell lysis
   d) Stimulation by helper T. cells
   e) Originating in the spleen

INDEX NUMBER.....

11) The initial phase of viral infection is controlled by the following **except**;

a) IFN-alpha
b) Interleukin 12
c) IFN-beta
d) Macrophages
e) Natural killer cells

12) The alternative complement pathway can be:-

a) Initiated by the formation of antigen antibody reaction
b) Predominantly a non-antibody initiated pathway
c) Activated by factors such as cells.
d) Stimulated by haematopoietic cells
e) A potent mediator of apoptosis

13) Major histocompatibility II is analogous to presentation by;

a) CD_8+ T-lymphocytes
b) CD_3+ lymphocytes
c) CD_4+ T-lymphocytes
d) Monocytes
e) Lymphocytes

14) T-cells stimulate B-cells to mature into;

a) CD_4 cells
b) Plasma cells
c) Macrophages
d) Ciliated cells
e) Memory cells

15) The following statement(s) is/are true about primary immune response;

a) Antibody may be detected within hours of antigen stimulation
b) IgM produced is relatively of high titre.
c) IgM produced is relatively of low titre
d) Antibody produced is predominantly of IgG class
e) Antibody produced is predominantly of IgM class.

INDEX NUMBER.....

16) T-cell deficiencies leads to susceptibility to;

a) Allergies
b) Pyogenic infection
c) IgA and IgG deficiency
d) Lack of immune complex formation
e) Opportunistic infections

17) The probability of success in organ and tissue transplantation increase as a result of:-

a) Surgical technique
b) Cytotoxic antibodies
c) Histocompatibility testing
d) Complement activation
e) Immunosuppression

18) An oncogene is a gene that is associated with;

a) Apoptosis
b) Hyper proliferation of cells
c) TCR signal transduction
d) Cancer.
e) BCR signal transduction

19) Which of the following uses enzyme horseradish peroxidase (HRP);

a) Fluorescent antibody test
b) FACs
c) ELISA
d) Western blotting
e) Southern blotting

20) As a result of a bee string the victim had an immediate severe reaction and nearly died, this reaction was caused by:-

a) B-cells
b) T-cells
c) Natural killer cells
d) Macrophages
e) Neutrophils

INDEX NUMBER.....

21) The following is/are responsible for repeated infection with influenza virus
   a) Antigenic shift
   b) Low stimulation of CD_4 + cells to produce memory cells
   c) Intracellular localization disable stimulation of immune system
   d) Neutralization of antibodies
   e) Antigenic drift

22) Which of the following are attributes of a good vaccine;
   a) Should be able to give short term protection
   b) Should not revert to virulence
   c) Should be made of expensive constituents
   d) Should be given in boosters
   e) Should be able to illicit humoral and cellular immune responses

23) Which of the following conditions develop as a result of allergic response;
   a) Arthus reaction
   b) Allergy
   c) Serum disease
   d) Myasthenia gravis
   e) Thyroiditis

24) The following are consistent with secondary immunodeficiency **except;**
   a) Genetic disorder
   b) Malnutrition
   c) HIV infection
   d) Immunosuppressive therapies
   e) Frequent exposure to gamma rays

25) The following test(s) is/are done in an immunology laboratory **except;**
   a) Cell flow cytometry
   b) Immunohistochemistry assay
   c) Radioimmunoassay
   d) Fungal infection assay
   e) Haemagglutination assay

INDEX NUMBER.....

26) An immunologic adjuvant is a substance that;

a) Reduces the toxicity of the immunogen
b) Enhances the immunogenicity of hapten
c) Enhances haematopoiesis
d) Enhances immune response against the immunogen
e) Enhances immunologic cross-reactivity

27) The following is/are disease(s) associated with T-lymphocytes deficiency;

a) Selective IgA deficiency
b) X-linked agammaglobulinaemia
c) DiGeorge syndrome
d) Common variable immunodeficiency
e) X-linked hyper IgM syndrome

28) MHC class II molecule;

a) Are found on all cells
b) Usually display self-antigen
c) Are found on antigen presenting cells
d) Cannot combine with other antigen
e) Are cell-surface glycoproteins that display antigens

29) The third line of defense involve;

a) Phagocytic white cells
b) Antimicrobial proteins
c) Inflammatory response
d) Lymphocytes
e) Antibodies

INDEX NUMBER.....

SECTION II (ESSAY 60 MARKS)

1) Describe inflammation process as an immune response.

(20 marks)

2)

a) Compare B-lymphocytes and T-lymphocytes. (10 marks)

b) State five (5) factors that can influence tolerance induction.

(10 marks)

3)

a) List five properties of killed virus vaccine.

b) Outline five advantages of a rapid diagnostic kit.

c) Explain type IV hyper sensitivity.

d) State five auto immune diseases.

(20 marks)

Immunology marking key

February, 2018

MCQ

1. D
2. D
3. A
4. C
5. E
6. B
7. B, C
8. B
9. E, D
10. D
11. D
12. B, C
13. C
14. B
15. B, E
16. B, E
17. C, E
18. D, B
19. C D
20. A
21. A, D
22. B, E
23. B
24. A
25. D
26. B
27. C, E
28. C, E
29. D, E

Essay 60 marks

1. Describe inflammation process
   a) Def: Inflammation is anon specific vascular and cellular response to trauma (1 mark)
   b) Purpose of inflammation
   i) To initiate the healing of the injured tissue (1 mark)
   ii) Protects the body by localizing and removing the injuring agents (1 mark)
   c) Causes of inflammation
      - Infectious micro-organism
      - Trauma
      - Surgery
      - Corrosive chemicals
      - Extreme heat and cold
      - Schemic damage to the body tissues

(1mark for any three total 3 marks)

d) Types of inflammation

i) **Acute phase** - takes 7-10 days (1mark), it is self-limiting and takes short duration

ii) **Chronic phase** - Involves formation of granuloma (1mark), may last for wee months or years. Involves infiltration of mononuclear cells and lymphocytes

e) Inflammatory process

i) **Vascular changes** (1 mark)

Involve an increase in capillary permeability that occurs 4 - 24 hours after injur

Dilation of blood vessels occurs and swelling (edema) is seen

ii) **Cellular changes** (1 mark)

Involve movement of phagocytic cells into the area of injury, cells involved include granulocytes, basophils, eosinophils, mononuclear cells (Macrophages). The movement of cells involved is dictated by inflammatory mediators like chemokines.

f) The cardical signs/Hallmark of inflammation

i) Redness (rubor) (1 mark)

Caused by blood vessels dilation, chemical mediators involved are histamine, serotonin, Bradykinins, prostaglandins which promote the vessel dilation (1 mark)

ii) Swelling (Tumor) (1mark)

Eodema fluid varies with the stage of inflammation (1mark)

iii) Pain (Bolar) (1mark)

Results from irritation of nerve ending by physical or chemical factors (1mark)

iv) Warm (calor) (1Mark)

The result of chemical activity and increased blood in the injured area (1mark)

v) Loss of function (Loss rom (1mark)

May occur due to pain causing reflex guarding or muscle spasm (1mark)

2a) Compare B-lymphocytes and T-lymphocytes (10marks)

	B - LYMPHOCYTES	T - LYMPHOCYTES
i)	Produced and mature in bone marrow (1 mark)	Produced in bone marrow and mature in thymus (1 mark)
ii)	Mediate humoral immune response (1 mark)	Cell mediated immunity (1 mark)
iii)	CD $_{19}$ marker(1 mark)	CD_3 marker (1 mark)

iv)	Maturation is antigen dependent (1 mark)	Maturation is antigen independent (1 mark)
v)	Recognize free antigens (1 mark)	Recognize antigen presented by antigen presenting cells in association with MHC (1 mark)

2b) State five (5) factors that can influence tolerance induction

i) Immunologic maturity of the host - neonates are immunologically immature and well accept allograft that would be rejected by mature host (2 marks)

ii) Structure of antigen - simple molecules induce tolerance more readily than complex ones (2 marks)

iii) Dose of antigen - Very high and very low doses of antigens may result in tolerance (2 marks)

iv) T-cells become tolerant more readily and remain tolerant longer than B-cells (2 marks)

v) The continuous presence of antigen helps to maintain tolerance (2 marks)

vi) Administration of immunosuppressive drugs enhances tolerance as in transplantation (2 marks)

3a) List five properties of killed virus vaccine

- It is safe and cannot revert to virulence (1 mark)
- Does not require cold chain for storage (1 mark)
- Expensive to prepare (1 mark)
- Des not illicit humoral and cellular immune response (1 mark)
- Requires boosters (1 mark)
- Majority are administered through infection (1 mark)

3b) Outline five advantages of a rapid diagnostic kit

i) Easy to use (1 mark)

ii) It takes short duration of time to get results (1 mark)

iii) They are cheap compared to molecular techniques (1 mark)

iv) Can be used in rural set up where electricity is absent (1 mark)

v) It is highly specific (1 mark)

vi) Chances of contamination are reduced as compared to molecular technique (1 mark)

01 19
16
41
99
073

3c) Explain type IV hypersensitivity

i) Type IV hypersensitivity is also known as cell mediated or delayed type hypersensitivity (1 mark)

ii) The examples of type IV hypersensitivity include contact dermatitis, contact rhinitis, tuberculin (1 mark)

iii) Type IV hypersensitivity takes 24 - 48 hours. (1 mark)

iv) No histamine or chemically related substances are released from cells (1 mark)

v) Type IV hypersensitivity is involved in pathogenesis of many autoimmune and infectious diseases i.e. tuberculosis, leprosy, blastomycosis, histoplasmosis, toxoplasmosis and leishmaniasis (1 mark)

vi) Antigens involved in type IV hypersensitivity are cell-associated antigens (1 mark)

3d) State five autoimmune diseases

i) Hashimoto's disease of thyroid (1 mark)

ii) Sjogreus disease (1 mark)

iii) Sympathetic opthalmia (1 mark)

iv) Multiple sclerosis (1 mark)

v) Peripheral neurotitis (1 mark)

vi) Male sterility (1 mark)

vii) Rheumatoid arthritis (1 mark)

viii) Acquired haemolytic anaemia (1 mark)

ix) Idiopathic thrombocytopenia purpura (1 mark)